

Junior Makers · Class 7

# Temperature-Controlled Smart Fan

Hotter room, faster fan. All by itself!

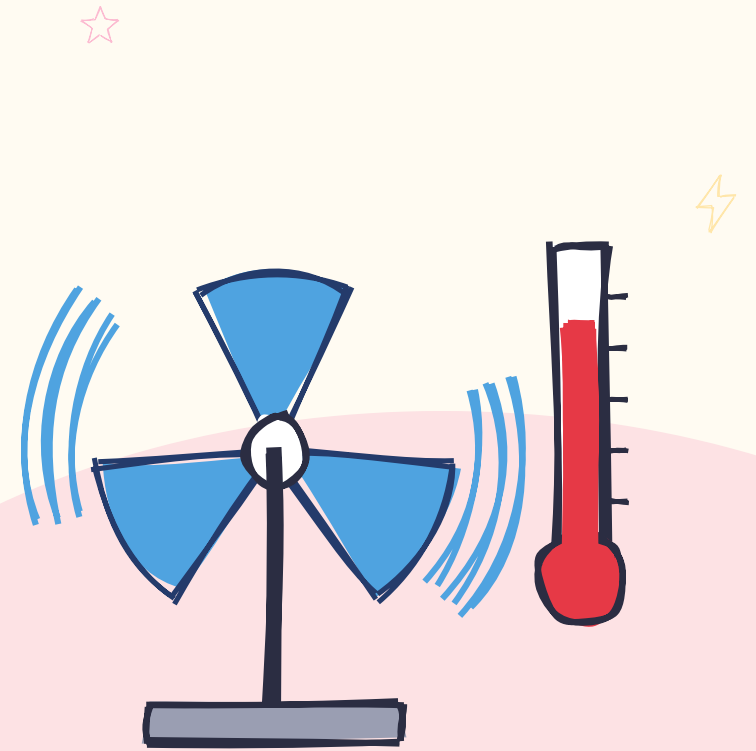
DHT22 sensing

L298N motor driver

Speed control

Did you know?

A fan does not make the air colder. It cools YOU by blowing heat and sweat away from your skin.





Let's think...

# Who switches the fan to full speed?

## A summer problem

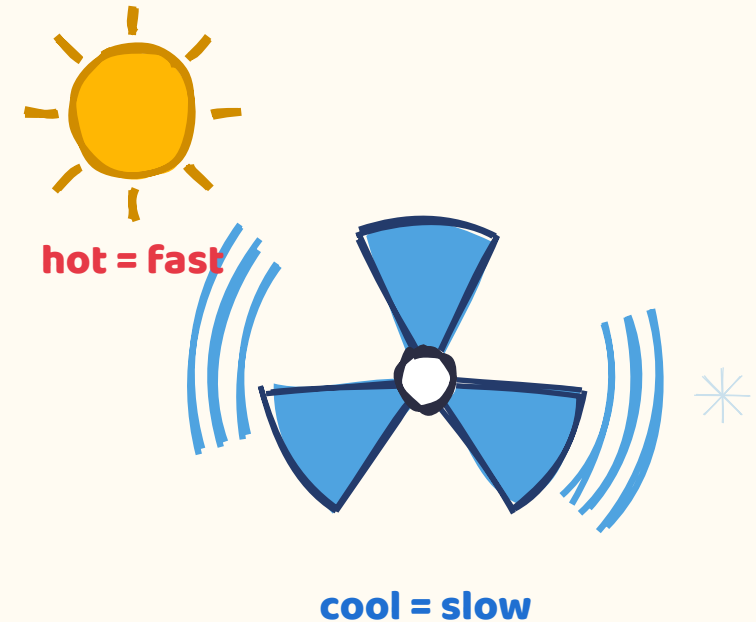
In June the afternoon is **very hot**, so the fan runs at full speed. At night it gets cooler, but nobody turns the fan down. It keeps using **electricity** we do not need.

Set a reminder?

Turn it off?

A fan that reads the temperature!

**Today:** we build a fan that measures the room and picks its **own speed**.





## Learning aims

# Today I will learn to...



1

Read the temperature with a **DHT22** sensor

2

Explain why a motor needs a **motor driver**

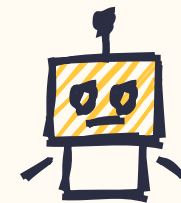
3

Change motor speed with **PWM** (0 to 255)

4

**Map** a temperature range to a speed range

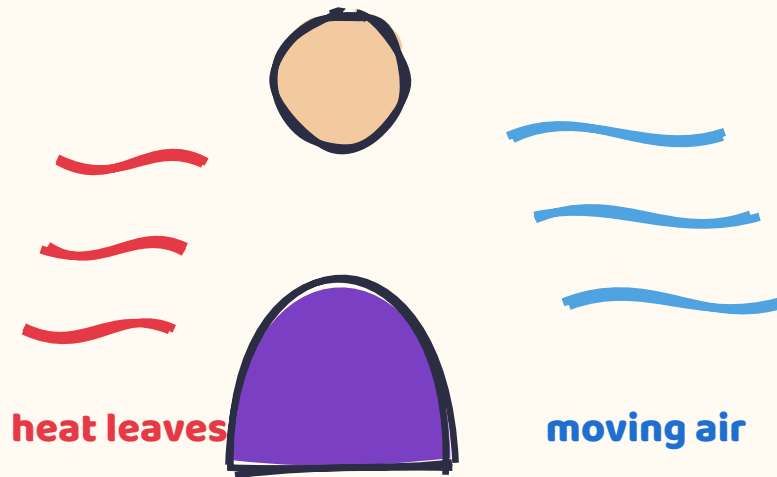
5

Add a **manual mode** with a knob**Keywords**temperature · DC motor · driver ·  
PWM · map · automatic · manual



## The science

# How does a fan cool us?



## Blowing heat away

Our body warms a thin layer of air next to our skin. A fan **blows it away** and brings new air. It also helps **sweat dry**, and drying sweat takes heat with it.

## Fun fact

A fan in an **EMPTY** room does nothing useful. Switch it off when you leave!

## Faster = cooler feeling

More air speed takes more heat away. But a faster motor also uses **more electricity**.

## Try it

Wet your hand and blow on it. Then blow on a dry hand. Which feels cooler?





Real world

# Machines that cool themselves



## Laptops

The fan speeds up when you play games and the chip gets hot



## Cars

A radiator fan switches on when the engine gets too hot



## Greenhouses

Fans and windows open when plants get too warm



## Inverter ACs

Slow down when the room is cool, saving electricity



Listen to a laptop or a fridge. Can you hear its fan **speed up** or **switch on** by itself?

### Same pattern

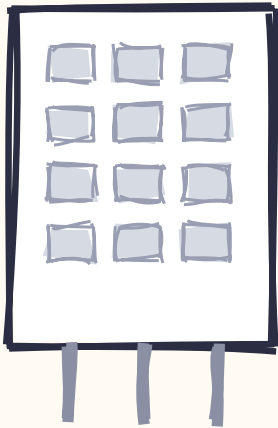
Sense the temperature, then decide how hard to cool. That is today's project!





Meet the sensor

# The DHT22: temperature and humidity



**DHT22**

## Fact file

Measures **temperature** in °C and **humidity** (how damp the air is).

Accurate to about **half a degree**.

Needs about **2 seconds** between readings.

## Warm it up

Hold the sensor gently between your fingers. The number rises in a few seconds.



| Pin | What it does  |
|-----|---------------|
| +   | 5V            |
| out | Data to pin 2 |
| -   | Ground        |

## Re-use!

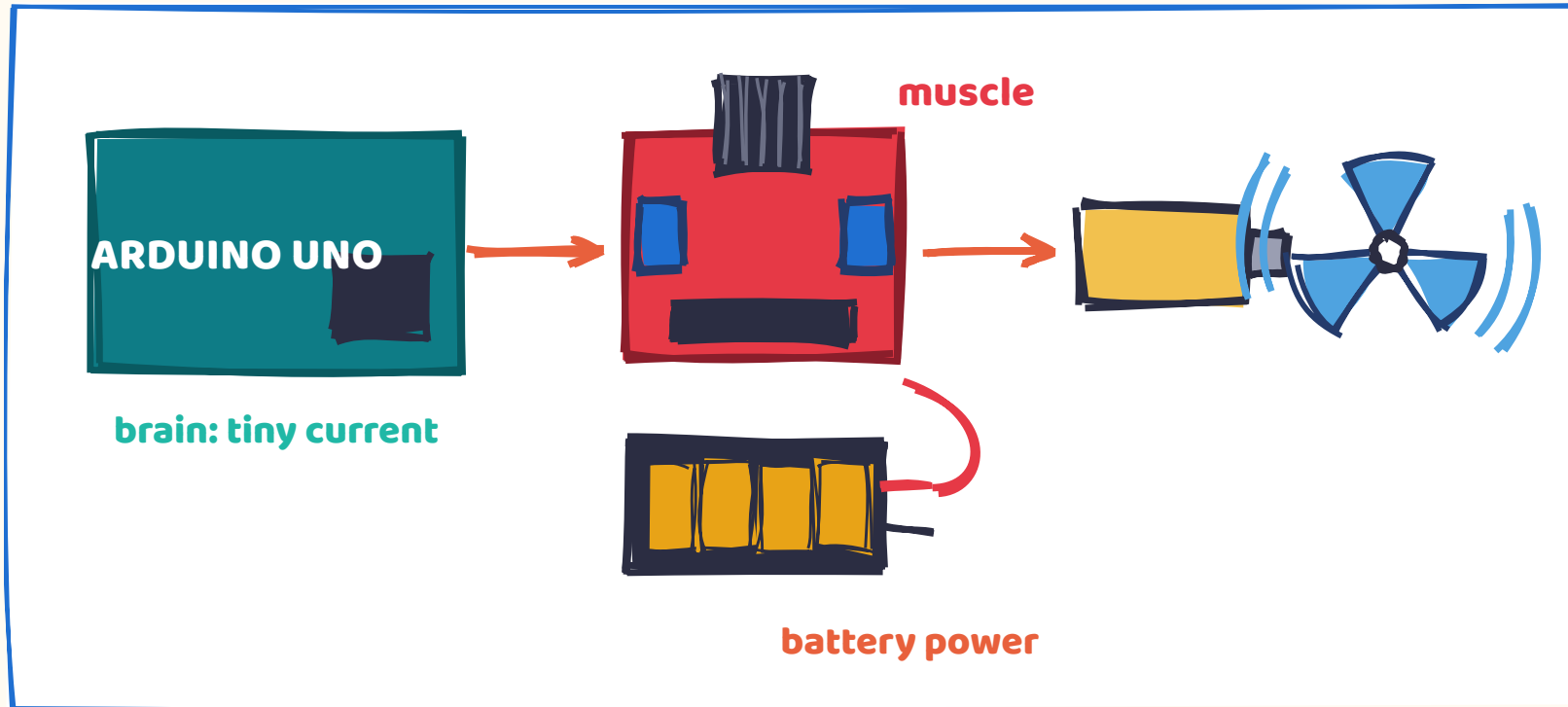
You met this sensor in the **Mini Weather Station**. Today it controls a machine!





Big idea

# Why can't the Arduino run the motor?



## Brain and muscle

An Arduino pin gives only a **tiny** current, like a drinking straw. A motor needs a **lot**, like a garden hose. Connect it straight to a pin and the Arduino could be **damaged**.

## The L298N

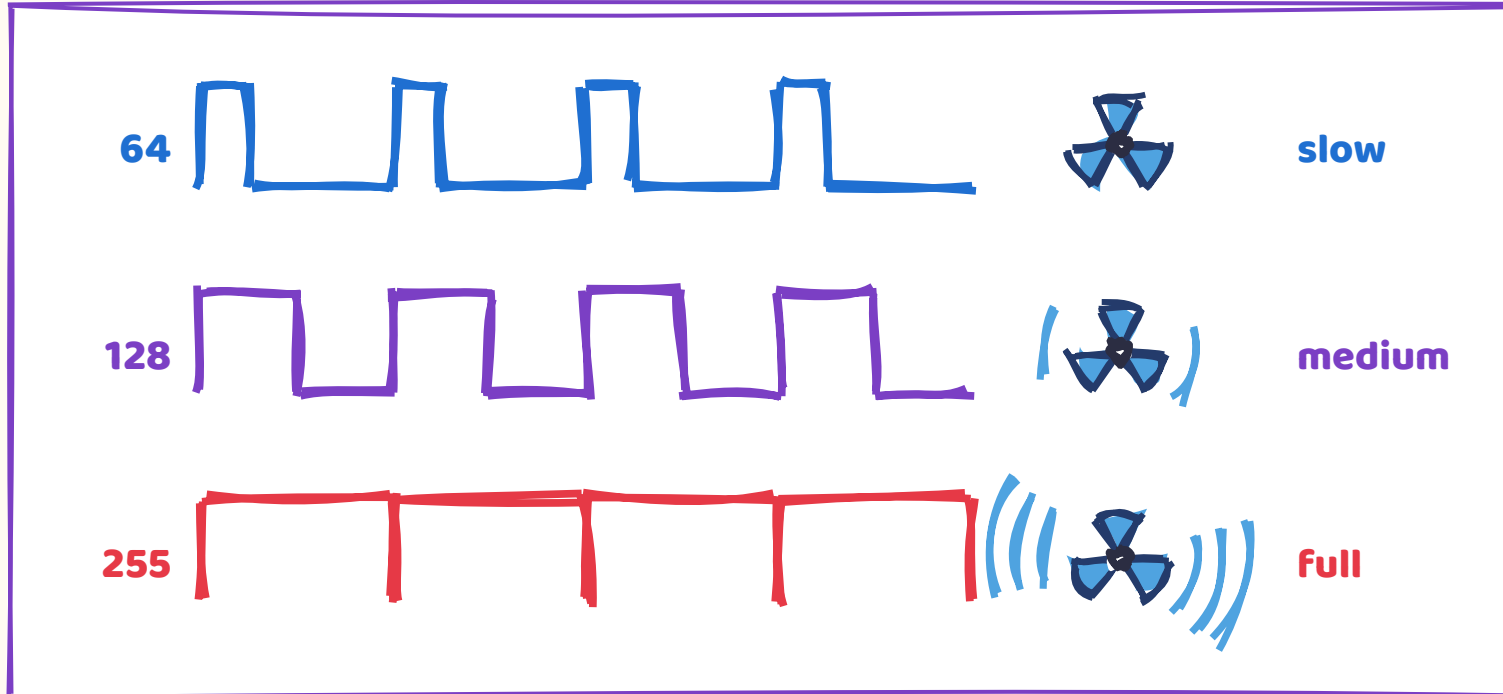
takes small signals from the Arduino and sends big power from the batteries.





Speed control

# PWM: blinking super fast



## How it works

The driver switches the motor **ON** and **OFF** hundreds of times a second. More ON time = more average power = **faster**.

We choose a number from **0** (stop) to **255** (full).

Hmm!

Below about 100 many motors just hum and do not turn. Test yours!





## Class game

# Be the smart fan!

### How to play

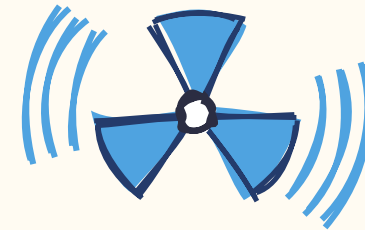
1. Everyone makes a **paper fan** (fold an A4 sheet).
2. The teacher calls out a temperature: "**22 degrees!**"
3. Below 25: **stop**. 25 to 35: wave **faster** as it rises. Above 35: **full speed!**
4. Wrong speed? You are out!

### Then discuss

How did you decide **how fast** to go for 30 degrees?  
That middle part is exactly what the Arduino has to work out.

### Big idea

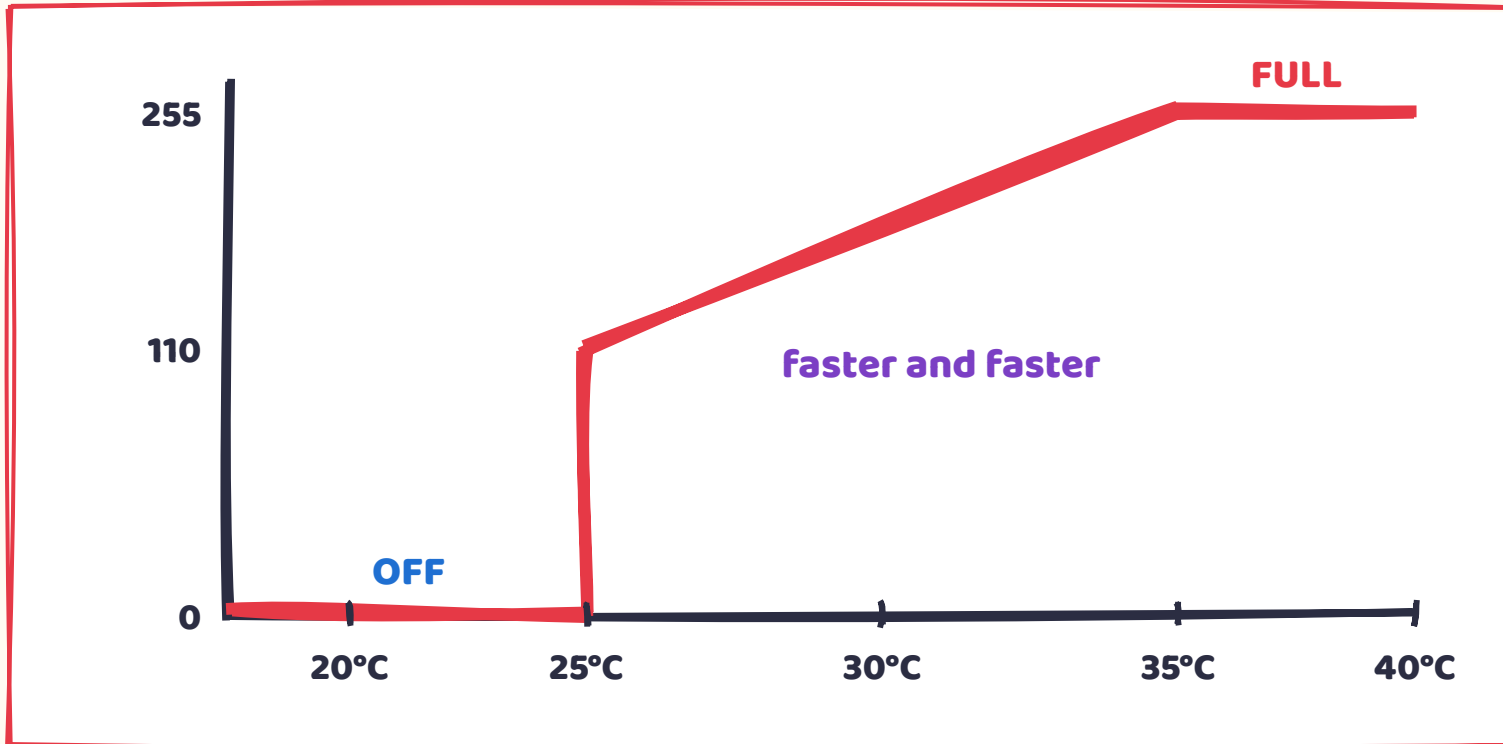
You just turned one number (temperature) into another number (speed). That is called **MAPPING**.





Mapping

# Turn temperature into speed



## The map block

```
map(temp, 25, 35, 110, 255)
```

25°C becomes speed 110

30°C becomes about 183

35°C becomes 255



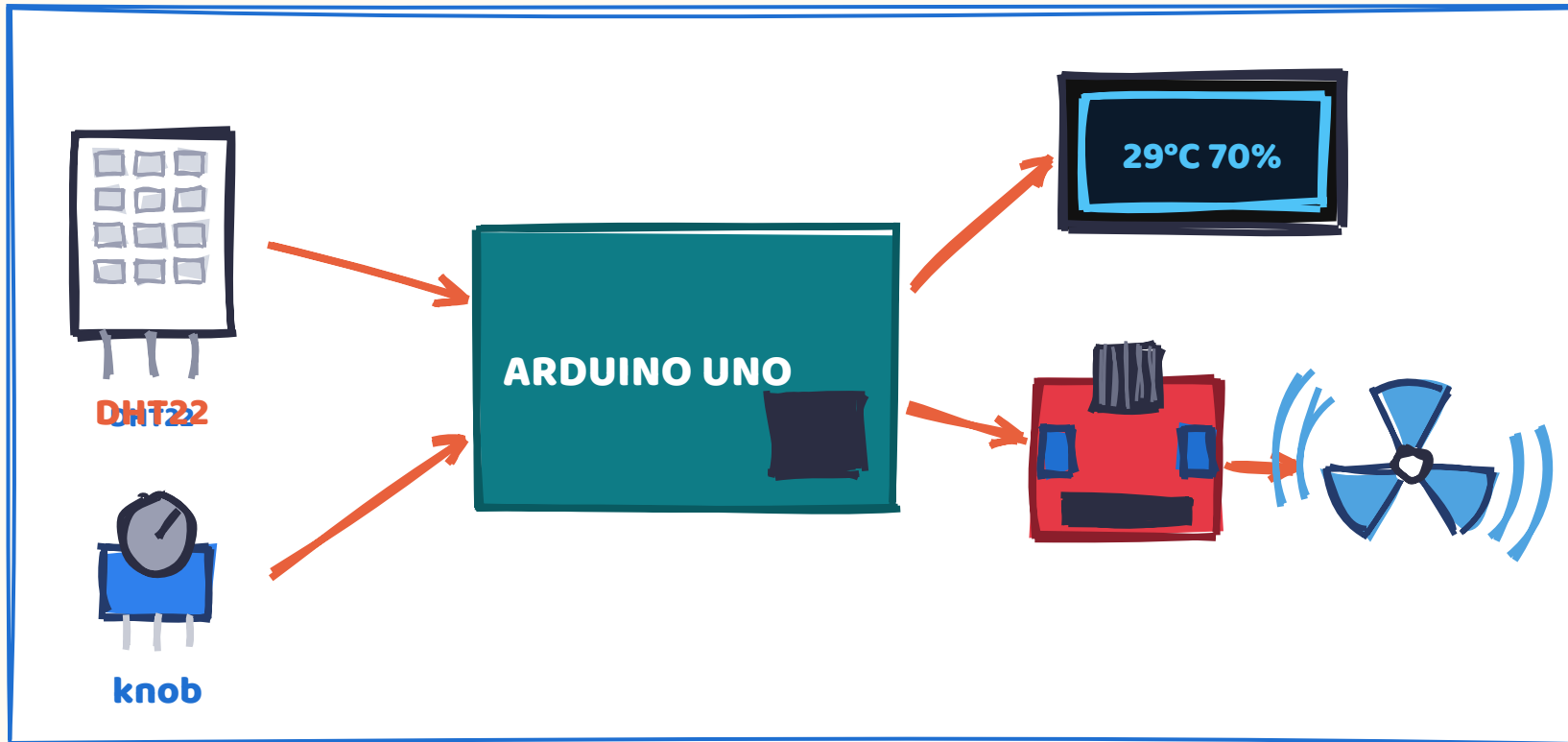
## Why start at 110?

Below that our motor would just hum. We skip the useless part!



The plan

# Sense, decide, spin



## OLED shows

Line 1: temperature

Line 2: fan speed in %

Line 3: AUTO or MANUAL

## Knob rule

Knob turned fully left = AUTO. Turn it up  
= you choose the speed.



Build it

# Our smart fan kit (per group)



**Arduino Uno**  
The brain



**DHT22 sensor**  
Feels the heat



**L298N driver**  
The muscle



**DC motor + blade**  
The fan



**10k knob**  
Manual control



**OLED 0.96 in**  
The display



Also: **4 × AA battery pack** for the motor, breadboard, jumper wires, USB cable, laptop, a card stand for the fan.

## Fan safety

Tie back long hair. Keep fingers away from the spinning blade!



Build it

# Wiring the smart fan

| Part              | Connect to            |
|-------------------|-----------------------|
| DHT22 + / out / - | 5V / pin 2 / GND      |
| OLED SDA / SCL    | A4 / A5               |
| Knob middle leg   | A0                    |
| L298N ENA         | pin 9 (remove jumper) |
| L298N IN1 / IN2   | pin 7 / pin 8         |
| L298N OUT1 / OUT2 | motor wires           |
| L298N 12V / GND   | battery + / -         |
| L298N GND         | also Arduino GND!     |

## Common ground!

Join the **battery minus**, the **L298N GND** and the **Arduino GND** together. Without it the signals have nothing to compare against and the motor will not listen.

## ENA jumper

Pull off the little black jumper on ENA so pin 9 can control the speed.



Code it

# Auto mode in ArduinoBlocks

**when Arduino starts**

set up OLED · pin 7 HIGH · pin 8 LOW (spin forward)

**forever**

set temp to DHT22 temperature (pin 2)

if temp &lt; 25

set speed to 0

else if temp &gt; 35

set speed to 255

else

set speed to map(temp, 25, 35, 110, 255)

analog write pin 9 → speed

OLED: temp and speed

wait 2000 ms

## Read it like a story

Cool? **Stop.**Very hot? **Full speed.**In between? **Map it.**

Then send the speed to pin 9.



## Classroom tip

Our room may already be 28°C! Read your room first, then set the start number 1 degree above it.





## Level 2

# Add manual mode with the knob



```
set knob to read analog A0
```

```
if knob < 50
```

```
  AUTO: use the temperature code
```

```
  OLED: AUTO
```

```
else
```

```
  set speed to map(knob, 50, 1023, 110, 255)
```

```
  OLED: MANUAL
```

## Why have manual mode?

People feel heat **differently**. Grandma may feel cold at 28°C while you feel hot. A good machine lets the **person** take control.

## Knob = map again!

The knob gives 0 to 1023. We map it to the speed range, just like temperature.



### Design idea

Put the knob where a user can reach it, away from the blade.





Test like an engineer

# Does the fan follow the heat?



| Test                           | Expect               | Temp    | Speed |
|--------------------------------|----------------------|---------|-------|
| Room air (no touching)         | slow or off          | ____ °C | ____  |
| Hold the DHT22 in your fingers | speeds up            | ____ °C | ____  |
| Let go and wait 1 minute       | slows down           | ____ °C | ____  |
| Knob fully left                | AUTO on OLED         | ____ °C | ____  |
| Knob half way                  | MANUAL, medium speed | ____ °C | ____  |

## Debug checklist

Motor hums only? Raise the lowest speed.

Motor never moves? Check common GND and the ENA jumper.

Temp shows "nan"? Check the DHT22 data wire on pin 2.



Wrong direction? Swap the two motor wires (or swap HIGH and LOW on pins 7 and 8).





Level up!

# Smart fan challenges

## Choose your challenge



**Heat light:** add an LED that glows red when it is above 35°C.



**Sticky air:** also speed up when humidity is above 70%.



**Speed bar:** draw a bar on the OLED that grows with the speed.



**Sleep timer:** turn the fan off after 30 minutes in MANUAL mode.

## Think like an engineer

If the temperature jumps between 24.9 and 25.1, the fan keeps switching on and off. How could we stop that?

## Boss level

Use TWO numbers: switch ON at 26, switch OFF at 24. Engineers call this hysteresis.



Show what you know

# What did we learn?

- ☒ A fan cools us by moving heat away from our skin
- ☒ The DHT22 measures temperature and humidity
- ☒ Motors need a driver: the Arduino is the brain, the L298N is the muscle
- ☒ PWM sets the speed from 0 to 255
- ☒ `map()` turns one range of numbers into another

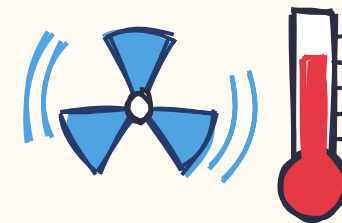


**Exit ticket:** why must we never connect a motor straight to an Arduino pin?



## At home

Find 3 machines at home with a fan inside. Does the fan speed change by itself?



**Next class:** a Sun-Tracking Solar Panel!

